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(1) Berhoulli Distribution :-

it is based on Discrete variable

→ {0,1} → Bernoulli trial Here $\boxed{n=1}$

→ $PMF = \begin{cases} p & \text{if } n=1 \\ q=1-p & \text{if } n=0 \end{cases}$

p: success
q: Failure

→ $P(X=r) = {}^nC_r \, p^r \cdot q^{n-r}$

* Mean = p
Variance = $p \cdot q \Rightarrow p(1-p)$

(2) Binomial Distribution :-

Here multiple Berhoulli trials

P.M.F $\boxed{P(X=r) = {}^nC_r \, p^r \cdot q^{n-r}}$

$$P(X=r) = \frac{n!}{(n-r)! \, r!} \, p^r (1-p)^{n-r}$$

Mean = np
variance = $npq \Rightarrow np(1-p)$

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